

A Review on Feature Selection and Feature Extraction for Text Classification

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Abstract—Day by day the number of text documents in digital form is increasing. Text classification is used to organize these text documents. However, text classification has the problem of high dimensionality of feature space. This high dimensionality of feature space is solved by feature selection and feature extraction methods and improves the performance of text categorization. The feature selection and feature extraction techniques remove the irrelevant features from the text documents and reduce the dimensionality of feature space. This paper presents the various feature selection and feature extraction methods. This paper also discusses various classifiers to classify the documents.

Keywords— *Text categorization; Feature selection; Feature extraction.*

I. INTRODUCTION

The text documents in digital form have drastically increased worldwide. Text classification has an important role to handle these documents. The main objective of text classification is it train the classifier on the basis of predefined categories [1,10]. The various application domain of Text classification generally includes web page classification, spam e-mail filtering, author identification, topic detection, etc [2].

Along with this, text classification has one limitation. This limitation includes high dimensionality of feature space due to the large number of features. This will increases the complexity of machine learning methods used for text classification and reduces the accuracy due to redundant or irrelevant terms in the feature space. To solve this problem, feature extraction and feature selection methods are used. [4,5]

Fig. 1 shows the flow of text classification. It includes four steps: document preprocessing, feature selection, feature extraction and text classification. For document preprocessing stop word removal, stemming, pruning of word technique, etc can be used. Feature selection can be defined as a process of selecting the subset from the original feature set on the basis of importance of features. There are three categories of feature selection methods: wrappers, filters and embedded methods [4,5]. Filters technique uses various scoring functionalities and select top-N features having the highest scores. They are computationally faster than wrapper method. The limitation is that the feature dependencies are not taken into consideration.

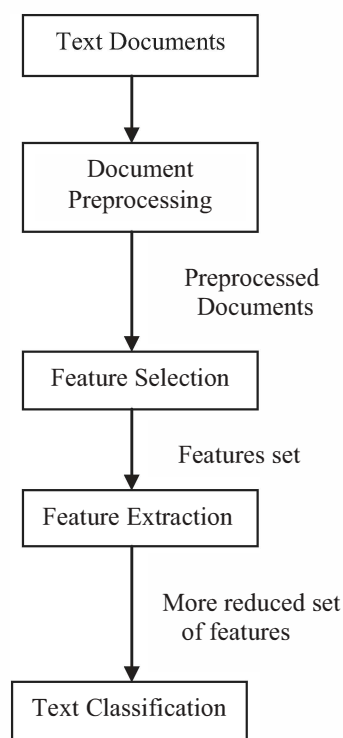


Fig. 1 Text classification flow [5]

Wrappers technique uses a learning model to evaluate the features, consider feature dependencies, it provide interaction between feature subset search and choice of the learning model. The limitation is that they are computationally slower than the filters. Embedded methods are same like the wrappers. However, they are computationally slower than the wrappers. In this paper filter methods are described.

Feature extraction can be defined as a process of extracting a set of new features from the features set that is generated in feature selection stage. Feature extraction methods include principal component analysis (PCA) [5], latent semantic indexing (LSI) [11], clustering methods, etc. Among these methods PCA is more powerful.

For text classification various classifiers such as decision Tree (DT), K Nearest Neighbor (KNN), Support Vector Machine (SVM), Naïve Bayes (NB) [4,9,10], etc. can be used.

Section-II presents the related work that describes the methods of feature selection, feature extraction and classifiers. Section-III presents the comparative analysis which shows advantage and disadvantage of all the methods.

II. RELATED WORKS

A. Feature selection

1) *Document Frequency (DF)* [8]: Document frequency is defined as the frequency of documents in which terms occur. For each term document frequency is calculated and if the document frequency of terms is less than some predetermined threshold then that term will be removed. DF thresholding can easily scale with large datasets with a computational complexity approximately linear to the number of training document.

1) *Information Gain (IG)* [4,8]: IG is the measure of how much information it contributes in the presence or absence of a term to make the classification decision on any class. IG reaches its maximum value when the document belongs to the respective class and the term is present in document. IG can be obtained using eq (1).

$$IG(t) = -\sum_{i=1}^M P(C_i) \log p(C_i) + p(t) + \sum_{i=1}^M p\left(\frac{C_i}{t}\right) \log p\left(\frac{C_i}{t}\right) + p(\bar{t}) \sum_{i=1}^M p(C_i/\bar{t}) \log p(C_i/\bar{t}) \quad (1)$$

2) *Gini Index (GI)* [4,8]: GI is simple feature selection method as it has simpler computation than the other methods. Gini Index can be calculated as

$$GI(t) = \sum_{i=1}^M p(t/C_i) p(C_i/t) \quad (2)$$

Where $P(t|C_i)$ is the probability of term t given presence of class C_i , $P(C_i|t)$ is the probability of class C_i given presence of term t , respectively.

3) *Chi square statistic* [4,8]: The goal of CHI2 test is to examine independence of two events i.e independence of class and term.

Two events A and B are said to be independent if

$$p(AB) = p(A)p(B) \quad (3)$$

We can calculate chi-square statistic using following equation

$$CHI2(t, c) = \sum_{t \in \{0,1\}} \sum_{c \in \{0,1\}} (N_{t,c} - E_{t,c})^2 / E_{t,c} \quad (4)$$

Where N indicates the observed frequency and E indicates the expected frequency for each state of term t and class c . CHI2 is a measure of how much expected counts E and observed counts N deviate from each other.

4) *Best Term (BT)* [6]: In this approach target class c is given and the BT algorithm will first finds the documents which belong to class c and then it will choose the features that predict the target class c with high precision. Next, it will find the documents which do not belong to target class c , but contain at least one of the features selected in above step. Then, in third step, a new feature set is generated by the combination of the two sets of features which are selected during the first and second step.

5) *Ambiguity Measure (AM)* [7]: Ambiguity Measure (AM) feature selection method will assign a higher score to the features which appear consistently in only one category. AM score is calculated for each feature. This method assigns score close to 1 if the feature is unambiguous; otherwise it assigns score close to 0. There is one threshold is given and based on this threshold, the features having AM score below that threshold that features are filtered and the features having AM score above that threshold are used for learning phase.

6) *Distinguishing Feature Selector (DFS)* [4]: Distinguishing Feature Selector method assigns higher scores to distinctive features and lowers scores to irrelevant features. In this method, ranking of features is done based on the following requirements:

a) A feature occurs in a single class and doesn't occurs in the other classes is irrelevant. So, it must assign a low score.

b) A feature rarely occurs in a single class and doesn't occur in the other classes is irrelevant. So, it must assign a low score.

c) A feature frequently occurs in all classes is irrelevant. So, it must assign a low score.

d) A feature occurs in some of the classes, is relatively distinctive. So, it must assign a relatively high score. Based on the first and second requirements, an initial scoring framework is constituted as

$$\sum_{i=1}^M \frac{p\left(\frac{C_i}{t}\right)}{p\left(\frac{t}{C_i}\right)} \quad (5)$$

Where M is the number of classes, $P(C_i|t)$ indicates the conditional probability of class C_i given presence of term t and is the conditional probability of absence of term t given class C_i , respectively. The above formula doesn't satisfy the third and fourth requirement. Therefore, the above formula is extended to

$$DFS(t) = \sum_{i=1}^M \frac{P(C_i/t)}{P\left(\frac{t}{C_i}\right) + P\left(\frac{t}{\bar{C}_i}\right) + 1} \quad (6)$$

Where, $P\left(\frac{t}{C_i}\right)$ is the conditional probability of term t given the classes other than C_i . Therefore, the third and fourth requirement is also satisfied. DFS assigns a score in between

0.5 and 1. It assigns score as 1 to the most significant features and 0.5 to the least significant features.

B. Feature extraction

1) *Principle Component Analysis (PCA)* [5]: Principle Component Analysis is dimension reduction technique. PCA extracts the information from a various datasets. The goal of PCA is to produce lower-dimensional feature set from the original dataset. In PCA it is very important to determine the number of principal components. The p number of principal components to be chosen among all of the principal components should be the principal components to represent the data at their very best. There are certain criteria in determining the optimal number of principal component such as broken-stick model, , cross-validation, Velicier's partial correlation procedure, Kaiser's criterion, Barlett's test for equality of eigen-values, Cattell's screen-test and cumulative percentage of variance.

2) *Latent Semantic Indexing (LSI)* [11]: Latent Semantic Indexing (LSI) is the dimension reduction technique. The goal of LSI is to projects the documents with latent semantic dimensions. The latent semantic space has fewer dimensions than the original space. Latent semantic indexing is relay on the application of Singular Value Decomposition to a matrix.

C. Classifiers

1) *Decision Tree (DT) Classifier* [4,9,10]: In Decision tree classifier classes are going to be rejected until the correct class is reached. To fulfill this condition, we have to split the feature space into unique regions corresponding to the classes. Binary classification tree is very simple and common decision trees. The purpose of this decision tree is to split the feature space into two different parts according to a specific threshold. A class is assigned with a feature vector along with a path of a decision tree. To design a classification tree, splitting criterion, class assignment rule and stop-splitting rule are going to be considered.

2) *K Nearest Neighbor (KNN) Classifier* [4,9,10]: The purpose of KNN classifier is to rank the document's neighbors among the training documents and to compare with the class labels of the k most similar neighbors. To measure the Similarity between two documents the cosine measure, Euclidean distance, etc. are used. The similarity score is used as the weight of the classes of the neighbor document. If any of the class is shared by more than one of the k-nearest neighbors, then the sum of the similarity scores of those neighbors is obtained from the weight of that particular shared class. However, it is difficult to find the optimal number of k.

2) *Naïve Bayes (NB) Classifier* [4,9,10]: Naïve bias method assumes the independence of feature i.e. one feature should independent from another feature under known priori

probability and class conditional probability. Training and testing phase in Naïve Bayes is easy for implementation and computation.

3) *Support Vector Machine (SVM) Classifier* [4,9,10]: Support vector Machine is supervised classification method. SVM algorithm consists of two types of versions: non-linear and linear versions. In non linear version, classes are not separated i.e. no straight lines can be found that separate the classes. In linear version, Classes are separated using the hyperplanes. Linear classifier is shown in fig. 2. Linear classifier is defined as $W^T x + B = 0$. Where, W is the direction of the hyperplane and B is the exact position of hyperplane.

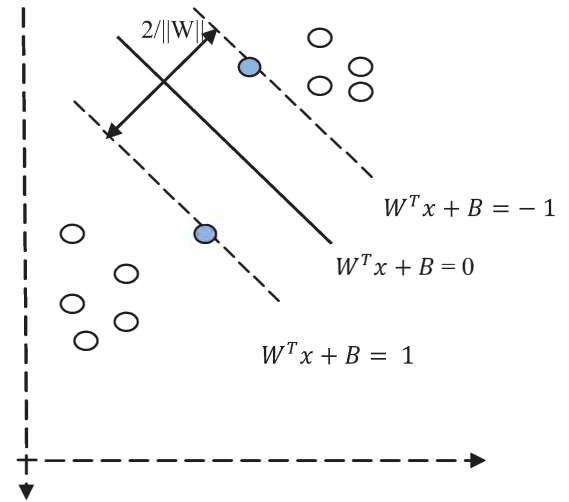


Fig. 2 Linear SVM classifier

SVM find this hyperplane by using margins and support vectors. The region between the two hyperplanes $W^T x + B = 1$ and $W^T x + B = -1$ is known as margin. $2/||w||$ is the width of the margin. The tuples that fall on the two hyperplanes are known as support vectors.

III. CONCLUSION

The text documents in digital form has drastically increased worldwide. Text classification has an important role to handle these documents. Text classification is the process of assigning predefined categories to the set of text documents. Text classification suffers from the problem of high dimensionality of feature space i.e it deals with the large number of features. To solve this problem, feature selection and feature extraction methods can be use as a preprocessing stage for text classification. In this study we have reviewed various feature selection and feature extraction methods as well as various classifiers.

III. COMPARATIVE ANALYSIS

A. Comparative analysis of feature selection methods

TABLE I. COMPARATIVE ANALYSIS OF FEATURE SELECTION METHODS

Method	Advantage	Disadvantage
Document Frequency	Simplest method with the lower cost in computation	Based on the assumption that the rare terms are non-informative, this contradicts with a principle of IR, where rare terms are more informative
Information Gain	Biased towards multi-valued attributes	Doesn't eliminate redundant features
Gini Index	Selects the features efficiently	Selects large number of features and doesn't eliminate redundant features
Chi-square Statistic	The value of chi-square is comparable with the same category terms	This method is not reliable for low frequency terms
Best term	The complexity of BT is linear with respect to training set documents and it is faster	Doesn't eliminate redundant features
Ambiguity measure	Assigns a higher score to the features which appear consistently in only one category	Score of each feature must be calculated and according to the score, all features should be ranked
Distinguishing feature selector	Selects distinctive terms and eliminate irrelevant terms	It achieves good accuracy as compared to above methods, however, the classification accuracy is still low

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B. Comparative analysis of feature extraction method

TABLE II. COMPARATIVE ANALYSIS OF FEATURE EXTRACTION METHODS

Method	Advantage	Disadvantage
PCA	Simplicity of technique and robustness in approximating the covariance or correlation matrix	The covariance matrix cannot be evaluated in an accurate manner
LSI	Easy to implement, understand and use.	It is a linear model, so not the best solution to handle non linear dependencies

C. Comparative analysis of classifiers

TABLE III. COMPARATIVE ANALYSIS OF CLASSIFIERS

Method	Advantage	Disadvantage
Decision Tree	Easily pick the best feature from the set of data	Overfitting is the problem and not so much accurate
K Nearest Neighbor	It is robust to noisy training data	Classification time is too long and it is difficult to find the optimal value of k
Naïve Bayes	Easy to implement and it requires less amount of training data	The assumption of independence of the class results in loss of accuracy
Support Vector Machine	Robust and very accurate	High algorithmic complexity and extensive memory requirements for large scale task

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